ARMA(p, q) Multivariate Normal Distribution

$$X_t - \mu - \phi_1 (X_{t-1} - \mu) - \dots - \phi_p (X_{t-p} - \mu)$$
$$= Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q} \qquad (+)$$

Assume $\underline{Z} = \begin{pmatrix} Z_t \\ Z_{t-1} \\ \vdots \\ Z_{t-q} \end{pmatrix} \sim N_{q+1}(\underline{0}, \sigma_Z^2 I)$

RHS of (+)

$$\cancel{\theta_1} Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}$$

Linear combination of $Z_t, \dots, Z_{t-q}$

$$(1, \theta_1, \dots, \theta_q) \begin{pmatrix} Z_t \\ \vdots \\ Z_{t-q} \end{pmatrix}$$

$$\underline{\theta}^T \underline{Z} \quad \text{where} \quad \underline{\theta} = \begin{pmatrix} 1 \\ \theta_1 \\ \vdots \\ \theta_q \end{pmatrix}$$

This has a normal distribution.

LHS of (+) is also normally distributed.

$$(X_t - \mu) - \phi_1 (X_{t-1} - \mu) - \dots - \phi_p (X_{t-p} - \mu)$$

has a normal distribution.

$$(1, -\phi_1, \ldots, -\phi_p)\begin{pmatrix} X_t - \mu \\ \vdots \\ X_{t-p} - \mu \end{pmatrix}$$

$$\underline{\phi}^T \underline{X} \qquad \underline{X} = \begin{pmatrix} X_t - \mu \\ \vdots \\ X_{t-p} - \mu \end{pmatrix}$$

$$\underline{\phi} = \begin{pmatrix} 1 \\ -\phi_1 \\ \vdots \\ -\phi_p \end{pmatrix}$$

$$\underline{\phi}^T \underline{X} = \underline{\theta}^T \underline{Z}$$

$$\Rightarrow \underline{X} = (\underline{\phi}^T)^{-1} \underline{\phi}^T \underline{\theta}^T \underline{Z}$$

$$\begin{pmatrix} X_t \\ \vdots \\ X_{t-p} \end{pmatrix} = \underline{\mu} + (\underline{\phi}^T)^{-1} \underline{\phi}^T \underline{\theta}^T \underline{Z}$$